Mathematical Methods for Physics—Lecture 20

Nonhomogeneous Boundary Conditions and Forced Problems

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We extend the finite-interval method of Lecture 19. First make the boundary conditions homogeneous, then expand in the corresponding spatial eigenfunctions. The modal amplitudes satisfy forced ODEs, solved by variation of constants or by the Laplace transform recalled from the earlier integral-transform unit. For a static source and static boundary data, a suitable stationary particular solution may simplify the PDE and the boundary conditions simultaneously.

1 Making Boundary Conditions Homogeneous

1.1 Subtract an auxiliary function

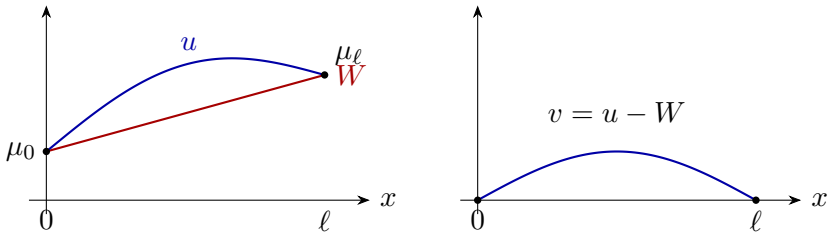
Suppose first that the endpoint data are constants. Choose a smooth $W = W(x)$ with the prescribed endpoint values or derivatives and write

$$v(x,t) = u(x,t) - W(x). \tag{1}$$

For example, if $u(0,t) = \mu_0$, $u(\ell,t) = \mu_\ell$, choose

$$W(x) = \mu_0 + \frac{\mu_\ell - \mu_0}{\ell} x. \tag{2}$$

Then $v(0,t) = v(\ell,t) = 0$.



Matching boundary data is the purpose of W; it need not satisfy the original PDE. Its residual must therefore be retained. For a time-independent W,

$$u_{tt} - a^2 u_{xx} = f \quad \Longrightarrow \quad v_{tt} - a^2 v_{xx} = f + a^2 W'', \tag{3}$$

$$u_t - \kappa u_{xx} = f \quad \Longrightarrow \quad v_t - \kappa v_{xx} = f + \kappa W''. \tag{4}$$

The new initial displacement is $\varphi - W$, and for the wave equation the initial velocity remains ψ. Thus making the boundary homogeneous is different from making the source vanish.

1.2 Four useful choices on an interval

Derivatives below are coordinate derivatives u_x, not outward normal derivatives.

1. **Two prescribed values:** $u(0,t) = \mu_0$, $u(\ell,t) = \mu_\ell$. Take $W = \mu_0 + (\mu_\ell - \mu_0)x/\ell$.
2. **Value on the left, derivative on the right:** $u(0,t) = \mu_0$, $u_x(\ell,t) = \nu_\ell$. Take $W = \mu_0 + \nu_\ell x$.
3. **Derivative on the left, value on the right:** $u_x(0,t) = \nu_0$, $u(\ell,t) = \mu_\ell$. Take $W = \nu_0(x - \ell) + \mu_\ell$.

4. **Two prescribed derivatives:** $u_x(0,t)=\nu_0$, $u_x(\ell,t)=\nu_\ell$. Interpolate the derivative first:

$$W'(x)=\nu_0+\frac{\nu_\ell-\nu_0}{\ell}x, \qquad W(x)=\nu_0 x+\frac{\nu_\ell-\nu_0}{2\ell}x^2. \tag{5}$$

These choices are not unique. Adding a function that satisfies the associated homogeneous endpoint conditions gives another admissible lifting.

1.3 Extension to time-dependent boundary data

If the endpoint data depend on t, use the same constructions with those time-dependent coefficients. For $v=u-W(x,t)$, the complete transformed wave problem has

$$v_{tt}-a^2v_{xx}=f-W_{tt}+a^2W_{xx}, \tag{6}$$
$$v(x,0)=\varphi(x)-W(x,0), \tag{7}$$
$$v_t(x,0)=\psi(x)-W_t(x,0). \tag{8}$$

For heat conduction the corresponding formulas are

$$v_t-\kappa v_{xx}=f-W_t+\kappa W_{xx}, \qquad v(x,0)=\varphi(x)-W(x,0). \tag{9}$$

This extension requires sufficiently regular endpoint data and compatibility with the initial data. It is essential to update the source and initial data as well as the boundary conditions.

2 A Forced Wave Equation with Homogeneous Endpoints

2.1 Expansion in spatial eigenfunctions

After boundary homogenization, consider

$$\begin{cases} u_{tt}-a^2u_{xx}=f(x,t), & 0<x<\ell,\ t>0,\\ u(0,t)=u(\ell,t)=0,\\ u(x,0)=\varphi(x), \qquad u_t(x,0)=\psi(x). \end{cases} \tag{10}$$

Here f,φ,ψ denote the data *after* any lifting and initial-data adjustment. Set $k_n=n\pi/\ell$ and $\omega_n=ak_n$. The complete sine basis gives

$$u(x,t)=\sum_{n=1}^{\infty}q_n(t)\sin(k_nx), \qquad f(x,t)=\sum_{n=1}^{\infty}f_n(t)\sin(k_nx), \tag{11}$$
$$\varphi(x)=\sum_{n=1}^{\infty}\varphi_n\sin(k_nx), \qquad \psi(x)=\sum_{n=1}^{\infty}\psi_n\sin(k_nx), \tag{12}$$

with

$$f_n(t)=\frac{2}{\ell}\int_0^\ell f(x,t)\sin(k_nx)\,\mathrm{d}x, \tag{13}$$
$$\varphi_n=\frac{2}{\ell}\int_0^\ell \varphi(x)\sin(k_nx)\,\mathrm{d}x, \qquad \psi_n=\frac{2}{\ell}\int_0^\ell \psi(x)\sin(k_nx)\,\mathrm{d}x. \tag{14}$$

Substitution and orthogonality reduce the PDE to independent modal ODEs:

$$\boxed{q_n''+\omega_n^2q_n=f_n(t), \qquad q_n(0)=\varphi_n, \quad q_n'(0)=\psi_n.} \tag{15}$$

Either of the following ODE methods may be used.

2.2 Variation of constants and the causal response

The homogeneous solutions are $\cos(\omega_n t)$ and $\sin(\omega_n t)$. Variation of constants gives the zero-initial-data particular solution

$$q_{n,f}(t) = \frac{1}{\omega_n}\int_0^t f_n(\tau)\sin(\omega_n(t-\tau))\,\mathrm{d}\tau. \tag{16}$$

Indeed, differentiation gives

$$q'_{n,f}(t) = \int_0^t f_n(\tau)\cos(\omega_n(t-\tau))\,\mathrm{d}\tau, \tag{17}$$

$$q''_{n,f}(t) = f_n(t) - \omega_n^2 q_{n,f}(t), \tag{18}$$

and both initial values of $q_{n,f}$ vanish. Adding the initial-data response,

$$q_n(t) = \varphi_n\cos(\omega_n t) + \frac{\psi_n}{\omega_n}\sin(\omega_n t) + \frac{1}{\omega_n}\int_0^t f_n(\tau)\sin(\omega_n(t-\tau))\,\mathrm{d}\tau. \tag{19}$$

The forcing acts through the causal kernel $\sin(\omega_n t)/\omega_n$.

2.3 The same formula from the Laplace transform

Recall the one-sided transform, for $\mathrm{Re}\,p$ sufficiently large,

$$Q_n(p) = \mathcal{L}[q_n](p) = \int_0^\infty q_n(t)e^{-pt}\,\mathrm{d}t, \qquad F_n(p) = \mathcal{L}[f_n](p). \tag{20}$$

The derivative identity includes both initial values:

$$\mathcal{L}[q_n''] = p^2 Q_n - p\varphi_n - \psi_n. \tag{21}$$

Transforming (15) gives

$$Q_n(p) = \frac{F_n(p) + p\varphi_n + \psi_n}{p^2+\omega_n^2}. \tag{22}$$

Use the inverse pairs

$$\mathcal{L}^{-1}\left[\frac{p}{p^2+\omega^2}\right] = \cos(\omega t), \qquad \mathcal{L}^{-1}\left[\frac{1}{p^2+\omega^2}\right] = \frac{\sin(\omega t)}{\omega}, \tag{23}$$

and the causal convolution theorem

$$(r *_t s)(t) = \int_0^t r(\tau)s(t-\tau)\,\mathrm{d}\tau, \qquad \mathcal{L}[r *_t s] = \mathcal{L}[r]\,\mathcal{L}[s]. \tag{24}$$

This reproduces (19). The first two terms represent initial displacement and velocity; the third is the forcing response. Substitute them into the spatial sine series to reconstruct $u(x,t)$.

Zero modes for other endpoint pairs. For a Neumann–Neumann pair, the constant spatial mode has zero frequency. Instead of dividing by $\omega_0 = 0$, solve it separately:

$$q_0'' = f_0(t), \qquad q_0(t) = \varphi_0 + \psi_0 t + \int_0^t (t-\tau)f_0(\tau)\,\mathrm{d}\tau. \tag{25}$$

Here f_0, φ_0, ψ_0 are spatial averages, with normalization $1/\ell$.

3 A Static Source and Nonzero Endpoint Values

3.1 Problem statement

Consider the problem recorded on textbook p.286:

$$\begin{cases} u_{tt} - a^2 u_{xx} = \sin(2\pi x/\ell)\cos(2\pi x/\ell), & 0 < x < \ell,\ t > 0, \\ u(0,t) = 3, \qquad u(\ell,t) = 0, \\ u(x,0) = 3(1 - x/\ell), \qquad u_t(x,0) = \sin(4\pi x/\ell). \end{cases} \tag{26}$$

Put $k = 4\pi/\ell$ and $\omega = ak$. The source is

$$\sin(2\pi x/\ell)\cos(2\pi x/\ell) = \tfrac{1}{2}\sin(kx). \tag{27}$$

Both the source and the boundary values are time independent. We can therefore look for one stationary particular solution that handles them together.

3.2 Main calculation: a stationary particular solution

Find $W = W(x)$ satisfying

$$-a^2 W'' = \tfrac{1}{2}\sin(kx), \qquad W(0) = 3, \quad W(\ell) = 0. \tag{28}$$

Integrating twice gives

$$W(x) = \frac{\sin(kx)}{2a^2k^2} + C_1 x + C_2. \tag{29}$$

Since $\sin(k\ell) = 0$, the endpoint conditions give $C_2 = 3$ and $C_1 = -3/\ell$:

$$\boxed{W(x) = 3\left(1 - \frac{x}{\ell}\right) + \frac{\sin(kx)}{2\omega^2}.} \tag{30}$$

Set $v = u - W$. The transformed problem is entirely homogeneous in its equation and endpoints:

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0, \\ v(0,t) = v(\ell,t) = 0, \\ v(x,0) = -\sin(kx)/(2\omega^2), \qquad v_t(x,0) = \sin(kx). \end{cases} \tag{31}$$

Only the fourth fixed-end sine mode is present. From Lecture 19,

$$v(x,t) = \left[-\frac{\cos(\omega t)}{2\omega^2} + \frac{\sin(\omega t)}{\omega}\right]\sin(kx). \tag{32}$$

Adding (30) gives the complete solution

$$\boxed{u(x,t) = 3\left(1 - \frac{x}{\ell}\right) + \left[\frac{1 - \cos(\omega t)}{2\omega^2} + \frac{\sin(\omega t)}{\omega}\right]\sin(kx).} \tag{33}$$

The stationary particular solution removes the static source and the nonzero endpoint data together; only one homogeneous vibration mode remains.

3.3 Alternative: boundary lifting followed by modal convolution

If we subtract only $L(x) = 3(1 - x/\ell)$, then $z = u - L$ has homogeneous endpoints but still satisfies

$$z_{tt} - a^2 z_{xx} = \tfrac{1}{2}\sin(kx), \qquad z(x,0) = 0, \quad z_t(x,0) = \sin(kx). \tag{34}$$

Writing $z = r(t)\sin(kx)$ gives

$$r'' + \omega^2 r = \tfrac{1}{2}, \qquad r(0) = 0, \quad r'(0) = 1. \tag{35}$$

Equation (19) yields

$$r(t) = \frac{\sin(\omega t)}{\omega} + \frac{1}{2\omega}\int_0^t \sin(\omega(t-\tau))\,\mathrm{d}\tau = \frac{\sin(\omega t)}{\omega} + \frac{1-\cos(\omega t)}{2\omega^2}. \tag{36}$$

Equivalently, its Laplace transform is

$$R(p) = \frac{1 + 1/(2p)}{p^2 + \omega^2}. \tag{37}$$

Both recover (33). This alternative illustrates the general forced-mode formula; the stationary choice of W is particularly convenient for this static example.

3.4 Check the reconstructed solution

Write $u = L + r(t)\sin(kx)$. Since $L'' = 0$, $\omega = ak$, and $r'' + \omega^2 r = 1/2$,

$$u_{tt} - a^2 u_{xx} = (r'' + \omega^2 r)\sin(kx) = \tfrac{1}{2}\sin(kx). \tag{38}$$

The sine vanishes at both endpoints, so the prescribed values 3 and 0 hold. Finally, $r(0) = 0$ and $r'(0) = 1$ give exactly the initial displacement and velocity.

Working Procedure

1. Identify the spatial interval, endpoint pair, source, and initial data.
2. Make the boundary conditions homogeneous and update the PDE and initial data. When a stationary particular solution also removes a static source, use that simplification.
3. Solve the spatial eigenvalue problem, including any zero mode.
4. Expand the source and initial data in that same basis.
5. Solve the modal ODEs by variation of constants or the Laplace transform, and reconstruct the physical field.
6. Verify the PDE, both boundary conditions, and all initial conditions.

The regularity of the data determines whether these expansions define classical solutions or must be interpreted in an integral or weak sense.

Textbook pointers. Nonhomogeneous boundary conditions: p.298. Forced eigenfunction expansion: p.282. The worked example above: p.286. Exercises: p.293, Problems 3(2),(3), 4(1), and 6(2),(4).